

## Title

Osmotic Laxative Adjunct to Multimodal Recovery Pathway Accelerates Postoperative Gastrointestinal Recovery After Hysterectomy: A Randomized Effectiveness Trial

## Abstract

Enhanced recovery pathways address multiple perioperative factors, yet bowel dysfunction remains common after abdominal surgery. We evaluated whether addition of an osmotic agent to a standardized enhanced recovery protocol improves gastrointestinal outcomes. We randomized 160 women (mean age 52 years) undergoing hysterectomy to receive either enhanced recovery care with oral osmotic laxative (n=80) or enhanced recovery care without additional intervention (n=80). The osmotic-supplemented group received laxative doses beginning immediately after surgery. Primary outcome was time to tolerance of regular diet, assessed via nursing assessment and patient report. Secondary outcomes included rates of clinically significant bowel dysfunction, time to first bowel movement, and patient-reported comfort. Median time to dietary advancement was significantly shorter in the osmotic group (20.8 hours vs. 29.1 hours;  $p<0.001$ ). Clinically significant bowel dysfunction occurred in 3.8% of the osmotic group versus 27.5% of the control group ( $p<0.001$ ). Time to flatus was reduced in the osmotic group (17.2 hours vs. 22.4 hours;  $p<0.001$ ). Patient satisfaction with recovery was higher in the osmotic-supplemented arm ( $p<0.001$ ).